

# Exploratory Analysis of Public Bus Stop Accessibility and Service Gaps Across Chennai

## Project Report

**Dataset:** Chennai bus stop audit   **Records:** 210   **Variables:** 85

### 1. Problem Definition

The project analyzes observed public bus stop conditions across Chennai to understand accessibility, passenger facilities, safety, information availability and infrastructure/service gaps. The emphasis is on exploratory analysis rather than causal inference.

### 2. Objectives

- Analyze pedestrian and wheelchair movement around bus stops.
- Assess bus-to-kerb boarding conditions.
- Evaluate shelter condition, roof, shade, seating and capacity.
- Identify encroachment, drainage and safety concerns.
- Assess stop-name and bus-route information availability.
- Explore land-use context and derive practical improvement priorities.

### 3. Methodology

The workflow consists of data loading, quality checks, whitespace/date standardization, duplicate removal, missing-value review, derived accessibility indicators, frequency analysis and visualizations. The pedestrian access score counts three Yes responses: usable space behind the shelter, usable space between shelter and road, and continuous footpath movement.

### 4. Key Findings

- Only 12.9% of audited stops satisfy all three pedestrian-access conditions simultaneously.
- 25.2% have usable space behind the shelter.
- 28.6% have usable space between the shelter and road.
- 26.2% allow pedestrians/wheelchair users to pass without stepping onto the road.
- Only 20.0% show a flush/good boarding position of less than 30 cm.
- 47.6% are partially encroached and 11.4% are heavily encroached.
- 27.1% show signs of water-logging or drainage issues.
- 17.6% report inadequate/no lighting, while 34.8% have no route information displayed.
- 27.6% have no clearly displayed stop name.

### 5. Recommendations

- Prioritize stops with poor pedestrian-access scores for footpath continuity, clear movement space and accessible approach paths.
- Reduce boarding gaps by improving kerb/platform design and aligning bus stopping positions.
- Remove or manage encroachments around high-risk stops, especially parked vehicles, vendors and construction material.
- Upgrade lighting and improve visibility at stops identified as unsafe or poorly lit.
- Provide clear, current bus route information and maintain legible stop-name signage.
- Address drainage and water-logging around affected stops.

- Review shelter capacity, shade, seating and roof condition where passenger demand exceeds available space.

## **6. Limitations**

The dataset represents audited observations and may not cover every bus stop in Chennai. Some fields are missing, and the survey includes subjective/observational responses. Therefore, results describe the supplied audit sample and should not automatically be generalized to the entire city.

## **7. Conclusion**

The analysis shows that accessibility and service-quality gaps are concentrated around pedestrian movement, boarding distance, encroachment, lighting and passenger information. A targeted improvement program using the derived indicators can help prioritize stops for infrastructure and service upgrades.